

Problem-Solution Fit

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Primary: University professors, academic evaluators, viva examiners Secondary: HR managers, technical interviewers, L&D trainers End Users: Students and job candidates being evaluated	6. CUSTOMER LIMITATIONS (BUDGET, DEVICES) [CL] Streamlit Cloud's 1 GB RAM ceiling limits model size CPU-only devices, no GPU access on the free tier Requires free/open-source tooling to stay within academic budget	5. AVAILABLE SOLUTIONS (PROS & CONS) [AS] Generic quiz apps: auto-graded but keyword-only, no voice input AI proctoring tools: video-focused, no semantic scoring Manual viva: flexible but subjective, unscalable, no audit trail
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Subjective grading, no semantic or acoustic analysis, time-consuming (8-12 students/hr), no audit trail, off-topic answers undetected - present in every oral assessment cycle	9. PROBLEM ROOT / CAUSE [RC] Manual oral assessment relies on each evaluator's subjective mental model, with no standardised, reproducible way to measure semantic accuracy or vocal delivery	7. BEHAVIOR + ITS INTENSITY [BE] Evaluators grow fatigued after 8-12 students/hour; scaling to 200 students requires 17-25 hours of continuous, high-intensity grading
3. TRIGGERS TO ACT [TR] Grade disputes with no evidence trail Cohort sizes outgrowing available evaluator hours Need for objective, reproducible scoring across sections	10. YOUR SOLUTION [SL] SkillEcho: an AI platform combining Whisper ASR, Bi-Encoder + Cross-Encoder NLP, Librosa DSP, and deterministic weighted scoring, with a downloadable PDF audit trail	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Streamlit Community Cloud web app
4. EMOTIONS BEFORE / AFTER [EM] Before: Frustration and fatigue for evaluators; anxiety over inconsistent grading for students After: Confidence in objective scoring; students receive a granular, structured metric breakdown		OFFLINE In-person oral exams and vivas that SkillEcho digitises and augments with automated scoring